

East West University

Department of Electrical and Electronic Engineering

Fall 2025, Final Examination

EEE 421 (RF and Microwave Engineering) Section 1 (DFP)

Full Marks: 100

Time: 70 Minutes

(Answer all the questions. The more work you show, the easier it is to give partial credit.)

1. Match a load impedance of $Z_L = 200 + j 150 \Omega$ to a 100Ω line using a single shunt open-circuited stub at 1 GHz. [30]

[CO2]

$$\begin{aligned} l &= 0.103 \lambda \\ d &= 0.165 \lambda \end{aligned}$$

2. Explain the difference between reactive near field and radiating near field of an antenna [10]

3. The radial component of the radiated power density of an antenna is given by – [30]

$$\overline{W_{rad}} = 10 \frac{\cos \varphi}{r^2} \hat{a}_r \text{ W/m}^2$$

Calculate -

- Total radiated power, P_{rad}
- Radiation Intensity, U

[CO5]

4. A lossless antenna has the following radiation intensity [30]

$$U = 10 \cos \theta$$

Calculate –

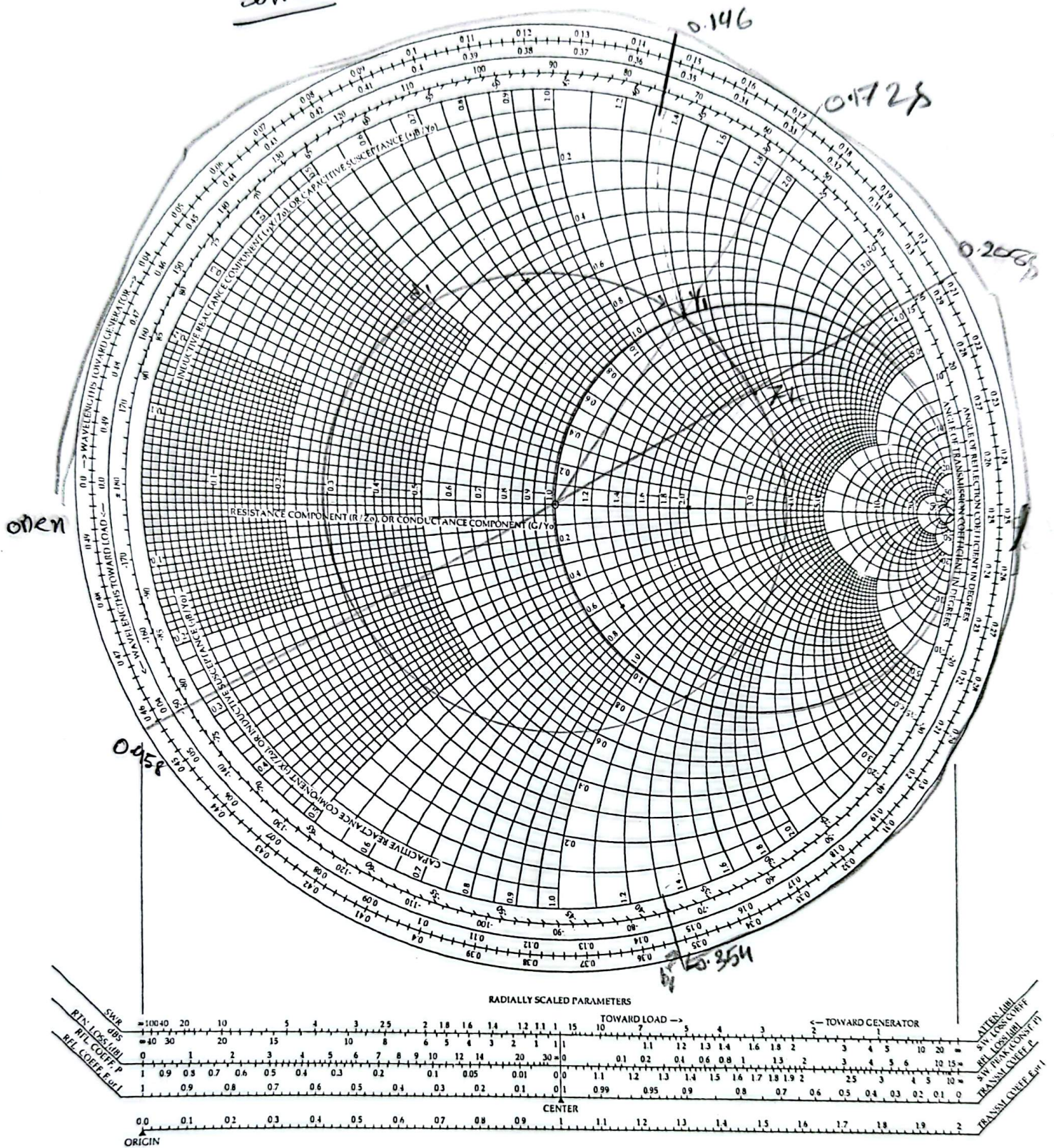
- Maximum directivity, D_0
- Maximum gain, G_0

[CO5]

Final
Solve

The Complete Smith Chart

Black Magic Design



$$Z_L = 1200 + j150$$

$$Z_0 = 100$$

$$Z_L = \frac{200 + j150}{100}$$

$$Z_L = 2 + j1.5$$

$$y_1 = 1 + j1.33$$

$$b_1 = -j1.33$$

$$d_2 = (0.5 - \overset{0.208}{\cancel{0.172}}) \delta + 0.172 \delta$$

$$= 0.468$$

$$L_2 = (\cancel{0.354 - 0.25}) \delta$$

$$= \cancel{0.104} \delta$$

$$L_2 = (0.5 - 0.354) \delta$$

$$= 0.146 \delta$$

$$d_1 = \cancel{(0.5 - 0.172)} \delta + 0.172 \delta$$

$$d_1 = (0.5 - y_L/2) \delta + y_L \delta$$

$$L_1 = (\text{center} - b_1) \delta$$

Final → 2025

2

2) Explain the difference between reactive near field and radiating near field of an antenna.

Soln

Reactive Near field: The region immediately surrounding the antenna where the reactive components of the electromagnetic field (stored energy) predominate. Energy oscillates back and forth between the antenna and the field without being radiated. Usually defined as $R < 0.62 \sqrt{D^3/\lambda}$

Radiating Near Field: The region between the reactive near field and the field, here, the radiation fields begin to dominate, but the angular distribution of the field (the radiation pattern) is still dependent on the distance from the antenna usually defined as $0.62 \sqrt{D^3/\lambda} < R < 2D^2/\lambda$

Q 3

The radial component of the power density of an antenna is given by

$$\vec{W}_{rad} = 10 \frac{\cos \phi}{r^2} \hat{r} \text{ W/m}^2$$

Calculate,

a) Total radiated power, P_{rad}

b) Radiation Intensity, U

Soln

a) Total radiated Power (P_{rad})

($0 \leq \theta \leq \pi/2$) and ($0 \leq \phi \leq \pi/2$) to keep $\cos \phi$

$$P_{rad} = \iint W_{rad} \cdot d\vec{s} = \int_0^{\pi/2} \int_0^{\pi/2} \left(10 \frac{\cos \phi}{r^2} \right) r^2 \sin \theta d\theta d\phi$$

$$P_{rad} = 10 \left[\int_0^{\pi/2} \sin \theta d\theta \right] \left[\int_0^{\pi/2} \cos \phi d\phi \right]$$

$$= 10 [-\cos \theta]_0^{\pi/2} [\sin \phi]_0^{\pi/2}$$

$$= 10(1)(1)$$

$$= 10 \text{ W}$$

b) Radiation Intensity (U)

$$U = r^2 W_{rad} = r^2 \left(10 \frac{\cos \phi}{r^2} \right) = 10 \cos \phi \text{ W/unit solid angle}$$

4

a) To

$$\vec{W}_{rad} = 20 \frac{\sin \theta}{r^2} \hat{r} \text{ W/m}^2$$

calculate:

a) Total radiated power, P_{rad} .

b) Radiation Intensity, U

$$P_{rad} = \int_0^{2\pi} \int_0^\pi U(\theta, \phi) \sin \theta d\theta d\phi$$

$$P_{rad} = \int_0^{2\pi} \int_0^\pi (20 \sin \theta) \sin \theta d\theta d\phi$$

$$P_{rad} = 20 \left[\int_0^{2\pi} d\phi \right] \left[\int_0^\pi \sin^2 \theta d\theta \right]$$

$$\int_0^{2\pi} d\phi = 2\pi$$

$$\int_0^\pi \sin^2 \theta d\theta = \frac{1 - \cos(2\theta)}{2}$$

$$\int_0^\pi \sin^2 \theta d\theta = \frac{1 - \cos(2\theta)}{2}$$

$$P_{rad} = 20 \left[\int_0^{2\pi} \frac{1 - \cos(2\theta)}{2} d\theta \right] \left[\int_0^{2\pi} d\phi \right]$$

$$= 20 [2\pi] \left[\frac{\theta}{2} - \frac{\sin(2\theta)}{4} \right]_0^\pi$$

$$= 20 \times 2\pi \times \frac{\pi}{2}$$

$$= 20\pi^2 = 197.39 \text{ W}$$

$$b) U = r^2 \left(\frac{20 \sin \theta}{r^2} \right)$$

$$= 20 \sin \theta$$

$$\vec{W}_{rad} = 20 \frac{\sin \phi}{r^2} \hat{r} \text{ W/m}^2$$

Calculate!

a) Total radiated power, P_{rad} .

b) Radiation Intensity, U

a) Total radiated power, P_{rad} .

$$P_{rad} = \int_0^\pi \int_0^\pi U \sin \theta d\theta d\phi$$

$$P_{rad} = \int_0^\pi \int_0^\pi (20 \sin \phi) \sin \theta d\theta d\phi$$

$$P_{rad} = 20 \left[\int_0^\pi \sin \theta d\theta \right] \left[\int_0^\pi \sin \phi d\phi \right]$$

$$P_{rad} = 20 [-\cos \theta]_0^\pi [-\cos \phi]_0^\pi$$

$$P_{rad} = 20 [-(-1) - (-1)] [-(-1) - (-1)]$$

$$= 20 \times 2 \times 2$$

$$= 80 \text{ W}$$

b)

$$U = r^2 \left(\frac{20 \sin \phi}{r^2} \right)$$

$$= 20 \sin \phi$$

Q4

A lossless antenna has the following radiation intensity

$$U = 10 \cos \theta$$

Calculate

a) Maximum directivity, D_0

b) Maximum gain, G_0

a) $U_{\max} \rightarrow 10 \cos \theta$ at $\theta = 0$

$$P_{\text{rad}} = \int_0^{2\pi} \int_0^{\pi/2} 10 \cos \theta \sin \theta d\theta d\phi = 10 (2\pi)$$
$$= 10 \int_0^{2\pi} d\phi \int_0^{\pi/2} \cos \theta \sin \theta d\theta$$
$$= 10 \cdot 2\pi \int_0^{\pi/2} \cos \theta \sin \theta d\theta$$
$$= 20\pi \int_0^{\pi/2} \frac{1}{2} \sin(2\theta) \rightarrow \sin(2\theta) = 2 \sin \theta \cos \theta$$
$$= \frac{20\pi}{2} \int_0^{\pi/2} \sin(2\theta) d\theta$$
$$= 10\pi \int_0^{\pi/2} \sin(2\theta) d\theta$$

$$D_0 = \frac{4\pi U_{\max}}{P_{\text{rad}}}$$
$$= \frac{4\pi \cdot 10}{10\pi}$$
$$= 4$$

$$= 10\pi \left[-\frac{\cos(2\theta)}{2} \right]_0^{\pi/2}$$
$$= 10\pi \left[-\frac{\cos(\pi)}{2} - \left(-\frac{1}{2}\right) \right]$$
$$= 10\pi \left(\frac{1}{2} + \frac{1}{2} \right)$$
$$= 10\pi$$
$$= 31.42$$

b) Maximum (a₀)

$$a_0 = \eta D_0 = 1 \times 4 = 4$$

$$a_0(\text{dB}) = 10 \log_{10}(4) = 6.02 \text{ dB}$$

Example 2.10

A lossless resonant half-wavelength dipole antenna, with input impedance of 73 ohms, is connected to a transmission line whose characteristic impedance is 50 ohms. Assuming that the pattern of the antenna is given approximately by

$$U = B_0 \sin^3 \theta$$

Soln

$$U_{\max} = U_{\min} = B_0$$

$$P_{\text{rad}} = \int_0^{2\pi} \int_0^\pi (B_0 \sin^3 \theta) \sin \theta d\theta d\phi$$

$$\Rightarrow P_{\text{rad}} = B_0 \int_0^{2\pi} d\phi \int_0^\pi \sin^4 \theta d\theta$$

$$= B_0 2\pi \int_0^\pi \left(\frac{1 - \cos 2\theta}{2} \right)^2 d\theta \rightarrow \sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$= B_0 2\pi \int_0^\pi \frac{1}{4} [1 - 2\cos 2\theta + \cos^2 2\theta] d\theta$$

$$\Rightarrow \frac{1}{4} B_0 2\pi \int_0^\pi \left[1 - 2\cos 2\theta + \frac{1 + \cos 4\theta}{2} \right] d\theta \rightarrow \cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$= 2\pi B_0 \int_0^\pi \left(\frac{1}{4} - \frac{1}{2} \cos 2\theta + \frac{1}{8} + \frac{1}{8} \cos 4\theta \right) d\theta$$

$$= 2\pi B_0 \int_0^\pi \left(\frac{3}{8} - \frac{1}{2} \cos 2\theta + \frac{1}{8} \cos 4\theta \right) d\theta$$

$$= 2\pi B_0 \left[\frac{3\pi}{8} - \frac{1}{2} \sin 2\theta + \frac{1}{8} \sin 4\theta \right]_0^\pi$$

$$= 2\pi B_0 \left(\frac{3\pi}{8} \right)$$

$$P_{rad} = B_0 \left(-\frac{3\pi^2}{4} \right)$$

Maximum Directivity

$$D_0 = 4\pi \frac{U_{max}}{P_{rad}}$$

$$\begin{aligned} D_0 &= 4\pi \frac{B_0}{B_0 \left(-\frac{3\pi^2}{4} \right)} \\ &= \frac{16\pi}{3\pi^2} \\ &= \frac{16}{3\pi} \\ &= 1.697 \end{aligned}$$

Maximum Absolute Gain G_0

The problem states the antenna is lossless, meaning it is lossless, meaning the radiation efficiency (e_{rad}) is 1.

$$G_0 = e_{rad} \times D_0 = 1 \times 1.697 = 1.697$$

$$G_0 (dB) = 10 \log_{10} (1.697) = 2.297 \text{ dB}$$